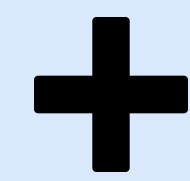
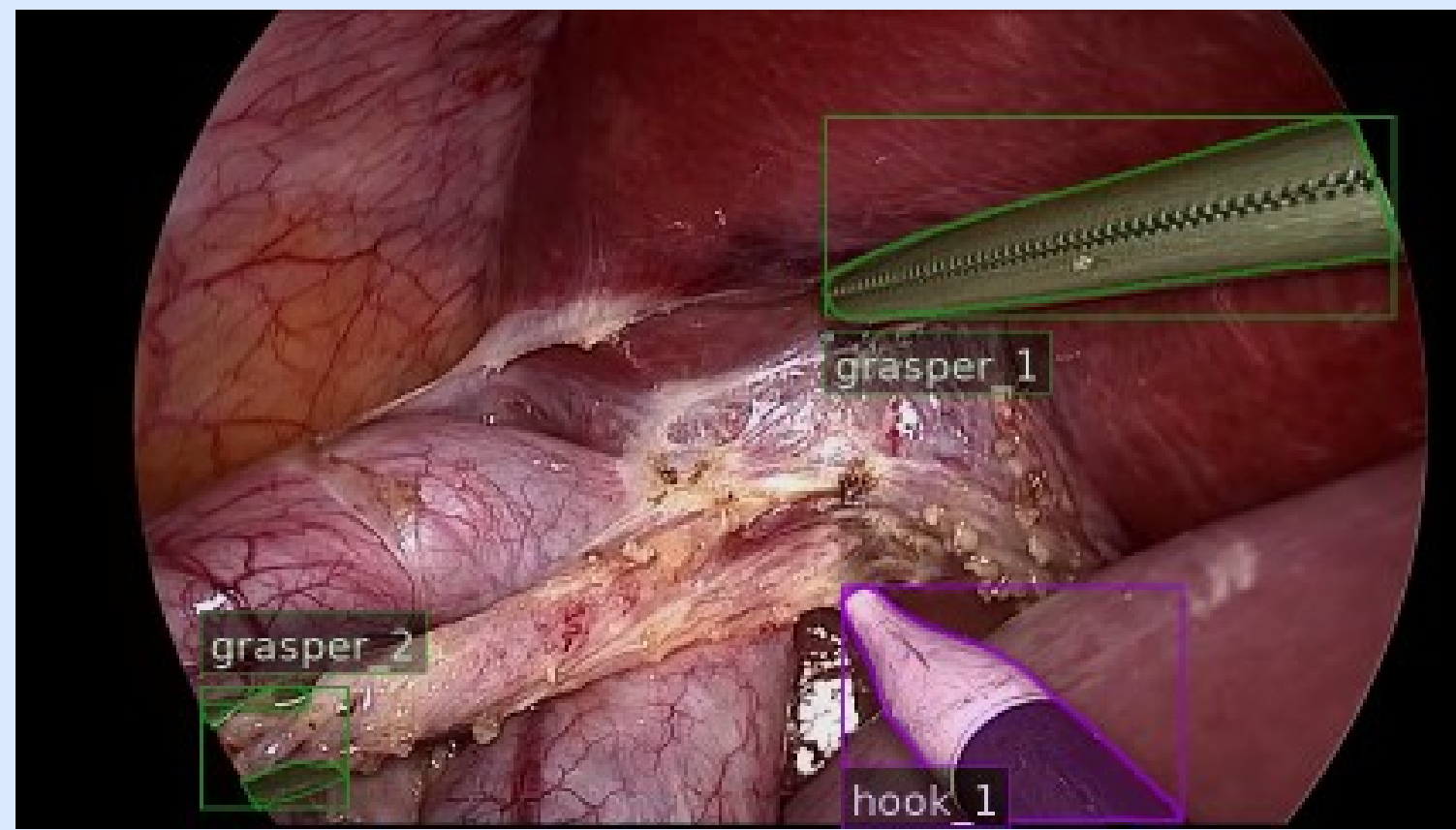


Teacher Rationale Generation

Inputs to teacher model

Instance Overlaid Surgical Image



Ground_truth

{<Grasper, retract, liver>,
....}

Instruction

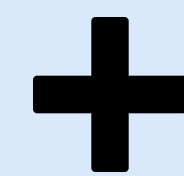
Analyse the laparoscopic image.
For each grounded instrument instance and its assigned triplet,
generate one concise reasoning trace that justifies the label.

Requirements:

- Use only visible evidence.
- Focus on the local instrument–tissue interaction.
- Keep all reasoning brief and discriminative.
- Return JSON only.

Generate:

- **contact_evidence**
- **action_evidence**
- **primary_target_hypothesis**
- **contrastive_target_hypothesis**
- **uncertainty**
- **final_triplet**



Generate reasoning supervision

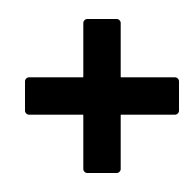
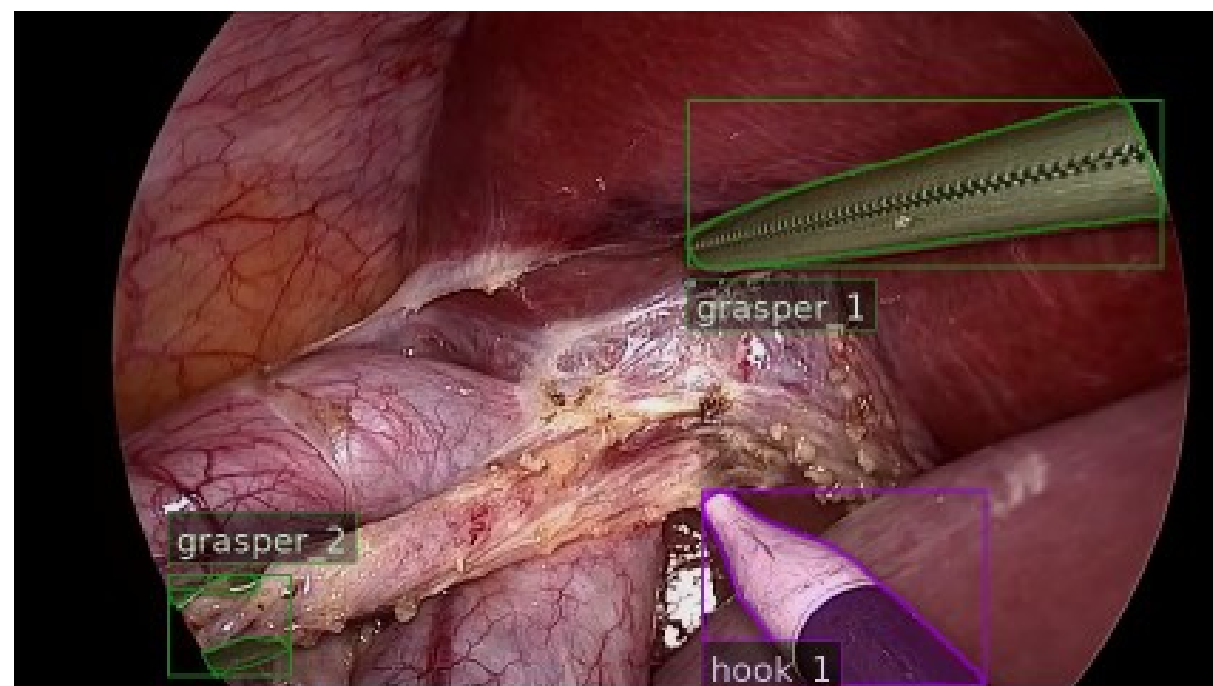


Outputs

```
{  
  "instance_id": "grasper_1",  
  "reasoning": {  
    "contact_evidence": [...],  
    "action_evidence": {...},  
    "primary_target": {...},  
    "contrastive_target": {...},  
    "uncertainty": "..."  
  },  
  "final_triplet": [  
    "Grasping Forceps",  
    "retract",  
    "liver"  
  ]  
}
```

Student Model Training

Inputs to student model



User:

Predict the contact
evidence, action
evidence,
primary_target,
contrastive_target, and
final_triplet..

Student model

Qwen3-VL-8B ❄️
+
LoRA 🔥

Pred

```
{  
  "instance_id": "grasper_1",  
  "reasoning": {  
    "contact_evidence": [...],  
    "action_evidence": {...},  
    "primary_target": {...},  
    "contrastive_target": {...},  
    "uncertainty": "..."  
  },  
  "final_triplet": [  
    "Grasping Forceps",  
    "retract",  
    "liver"  
  ]  
}
```

Reasoning supervision dataset